DRACO

Dotfile & Runtime Archive and Configuration Orchestrator

Quick Guide & Reference Card — v1.0.0

Portable encrypted backup for Linux workstations.
Dotfiles • SSH keys • KDE/GNOME config • Software manifests

Fedora 42+	Debian 12+	Ubuntu 24.04+	Arch Linux

License: GNU GPL v3 | Encryption: AES-256-CBC + PBKDF2 | Compression: zstd

1. WHAT IS DRACO?

DRACO backs up your **workstation identity**: shell configs, dotfiles, SSH/GPG keys, KDE or GNOME personalizations (including Konsole profiles and Spaceship prompt), and a complete list of installed software with a one-command reinstall script. It does **NOT** back up your personal data (Downloads, Documents, etc.).

All archives are **encrypted with AES-256-CBC** and compressed with **zstd**. Identical backups are deduplicated via SHA-256 (symlink instead of duplicate file). A full history with per-backup logs and diffs is maintained.

2. INSTALLATION

Dependencies (install first)

Fedora	`sudo dnf install tar openssl zstd newt`
Debian / Ubuntu	`sudo apt install tar openssl zstd whiptail`
Arch	`sudo pacman -S tar openssl zstd libnewt`

Install DRACO

```
tar xzf draco-step1.tar.gz
cd draco
bash install.sh
```

Note: Installs to ~/.local/bin/draco (user) or /usr/local/bin/draco (system).

3. FIRST RUN

```
draco
```

On first run, DRACO prompts for a **backup destination directory** and optionally installs an automatic schedule. All settings are saved to **~/.config/draco/draco.conf**.

■ NEVER store your encryption password in draco.conf. Enter it interactively, use DRACO_PASSWORD, or a password file (--password-file).

4. COMMAND REFERENCE

Command	Description
`draco`	Launch TUI (interactive mode)
`draco backup`	Run backup now
`draco restore`	Restore (interactive selection)
`draco restore <ID>`	Restore specific backup by ID
`draco list`	List all backups + sizes + dates
`draco log`	View log for last backup
`draco log <ID>`	View log for specific backup
`draco delete <ID>`	Delete a backup (with confirmation)
`draco schedule install`	Install automatic schedule
`draco schedule remove`	Remove automatic schedule
`draco schedule status`	Show schedule status + next run
`draco config`	Configuration wizard
`draco status`	Show config summary

5. OPTIONS & ENVIRONMENT

Option / Variable	Description
`-c FILE / --config FILE`	Use alternate config file
`--password-file FILE`	Read password from FILE (mode 600)
`-q / --quiet`	Suppress non-error output
`-v / --verbose`	Debug output
`--no-color`	Disable color output
`--version`	Print version
`-h / --help`	Show usage
`DRACO_PASSWORD=xxx`	Encryption password (env var, no prompt)

`DRACO_PASSWORD_FILE=path`	File with the password (mode 600)
`DRACO_CONFIG=path`	Override config file path
`DRACO_BACKUP_DIR=path`	Override backup directory
`DRACO_LOG_LEVEL=0-3`	0=DEBUG 1=INFO 2=WARN 3=ERROR
`DRACO_TUI_THEME=...`	ghost | blood | acid | matrix | void

6. AUTOMATED BACKUPS

Systemd user timer (recommended)

```
draco schedule install # choose systemd, daily, 02:00
```

The installer asks for the password once and stores it in **~/.config/draco/password** (mode 600, not included in backups).

Cron

```
draco schedule install # choose cron
```

```
# Or manually add to crontab:
```

```
0 2 * * * /home/user/.local/bin/draco backup -q --password-file
~/.config/draco/password
```

7. RESTORE WORKFLOW

1. Install DRACO on new machine (see Section 2)
2. Copy backup directory or mount backup storage
3. Set DRACO_BACKUP_DIR or run 'draco config'
4. Run 'draco restore' and select backup
5. Enter encryption password
6. Run the reinstall script:

   ```
   bash ~/.local/share/draco/restore-<ID>/reinstall.sh
   ```

7. Restart your session

■ KDE <-> GNOME restore: DE-specific settings are automatically skipped with a warning. All other configs (dotfiles, SSH, etc.) are restored normally.

8. RETENTION POLICIES

`all`	Keep everything (default). Disk grows indefinitely.
`daily`	Keep last N daily backups. Set DRACO_RETENTION_KEEP_DAILY.
`weekly`	Keep last N weekly backups.
`monthly`	Keep last N monthly backups.
`smart`	Best for long term: N daily + 1/week N weeks + 1/month N months + 1/year N years.

9. TUI THEMES

`ghost`	White on near-black (fg=255, bg=232). Native bash 256-color.

blood	Red on near-black (fg=196, bg=232).
acid	Yellow on near-black (fg=226, bg=232).
matrix	Green on near-black (fg=46, bg=232).
void	Cyan on near-black (fg=51, bg=232).

Set via: **draco config** or **DRACO_TUI_THEME=void draco**

10. KEY FILE LOCATIONS

~/.config/draco/draco.conf	Main configuration file
~/.local/share/draco/logs/	Session log files
$BACKUP_DIR/draco-YYYYMMDD-HHMMSS.*.enc	Encrypted archives
$BACKUP_DIR/.meta/*.meta	Per-backup metadata
$BACKUP_DIR/.meta/*.log	Per-backup log / diff
~/.config/systemd/user/draco-backup.*	Systemd units (if installed)
~/.local/share/draco/restore-<ID>/reinstall.sh	Generated reinstall script

11. SECURITY CHECKLIST

- Use a **strong password** (12+ chars, mixed). It protects SSH keys.
- **Never** store password in draco.conf. Use the prompt, DRACO_PASSWORD or a password file.
- Set backup directory permissions: **chmod 700 $DRACO_BACKUP_DIR**
- For unattended backups: **~/.config/draco/password** (chmod 600, created by draco schedule install)
- Back up your password somewhere safe. Without it, archives are unrecoverable.

12. KNOWN LIMITATIONS

- No user data backup (Downloads, Documents, media). By design.
- Monitor/display config excluded from restore (hardware-specific).
- KDE Activities not backed up.
- Flatpak: only overrides backed up, not per-app data.
- Reinstall script: packages renamed since backup will fail silently.
- npm/pip/snap lists are best-effort only.